



Fuzzy event-triggered control for nonlinear networked control systems

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Received 21 October 2021; received in revised form 16 January 2022; accepted 21 February 2022

Available online 26 February 2022

Abstract

This paper is focus on an event-triggered control design problem for nonlinear networked control systems with missing data and transmission delay in the interval type-2 (IT2) fuzzy form. An event-triggered controller is presented under a sampled-state-error mechanism. By dividing the event-triggered interval into some subsets, stability analysis is carried out based on Lyapunov-Krasovskii functional (LKF), and the stability of the closed-loop system is ensured. The proposed design is applied to the continuous stirred tank reactor system (CSTR) and the manipulator system. The control strategy is effective, and the merit of the event-triggered mechanism is indicated.

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1. Introduction

The control problem of nonlinear systems with parameter uncertainties is a challenging issue. Its controller design is more complex and difficult than that of nonlinear systems

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without parameter uncertainties. Thus, to effectively solve this problem, an interval type-2 (IT2) Takagi-Sugeno (T-S) fuzzy system [1] was constructed to model this class of systems. The authors first investigated the IT2 fuzzy control problem in [2], and then IT2 fuzzy control problems were investigated in [3–10] and references therein.

The controller is implemented by computers which transmit sampled-data signals. And, the sampled-data control design can provide a better theory guidance for engineering systems. There are some works which have reported IT2 fuzzy sampled-data control problems in [11–16]. In [11], the authors discussed the IT2 fuzzy $\mathcal{H}_\infty$ control problem, and the designed controller was of an ability of fault-tolerance and a resistance against external disturbances. In [12], the IT2 fuzzy stabilization problem was researched, and the sampling property was considered in the Lyapunov-Krasovskii functional (LKF). In [15], the IT2 fuzzy polynomial controller was presented, where the sum-of-square method was utilized to acquire the stability conditions.

In the signal transmission from computers, the unreliable transmission is inevitable. So, the networked control systems (NCSs) is an important area [17–25]. The IT2 fuzzy control was taken into account for unreliable NCSs with time delay in [22], where the unreliable communication obeys the Bernoulli random distribution, and the stochastic stability was ensured. In [23], a novel IT2 fuzzy fault detection scheme for NCSs was addressed, in which an augmented system was determined, and the asymptotical mean-square stability was acquired. The work [24] focused on an IT2 fuzzy stabilization problem for NCSs with stochastic transmission was addressed, and the exponential stability was guaranteed. The paper [25] was concerned with an IT2 fuzzy controller for NCSs with data dropout and transmission delay, and the $\mathcal{H}_\infty$ performance was ensured.

In comparison with the sampled-data control, the event-triggered control can further reduce the sampling number and save the limited bandwidth, which is attracting an extensive attention. In [26–35], the IT2 fuzzy event-triggered control problem was concerned. In [26], the authors put forward the IT2 fuzzy event-triggered controller for NCSs, and a dynamic output feedback was developed to achieve the $\mathcal{H}_\infty$ performance. In [27], the IT2 fuzzy event-triggered predictive control was considered for NCSs, and the work was an online MPC optimization problem. The paper [30] was devoted to the IT2 fuzzy polynomial event-triggered controller for NCSs, and the sufficient conditions in terms of the sum-of-square method was established. In [31], the asynchronous IT2 fuzzy event-triggered controller was discussed for NCSs subject to non-periodic DoS attacks, and a switched dynamical system was constructed. The authors proposed an IT2 fuzzy event-triggered sliding mode controller for NCSs with fading channels [34], and the input-to-state stability was utilized. These works display that the event-triggered control saves the implementation cost of the controller, where the number of transmitted data packets is decreased, and the occupancy rate of network resources is reduced. When missing data and transmission delay exist in NCSs simultaneously, there are a few reports on IT2 fuzzy event-triggered control. Thus, the requirement of engineering design can not be met, which motivates this study.

Based on the above analysis, the IT2 fuzzy event-triggered stabilization is concerned for NCSs. An IT2 fuzzy event-triggered controller under a sampled-state-error mechanism is designed. By using the LKF and dividing the holding interval of the zero-order-holder, the stability is conducted. The control design is applied to two systems, and the effectiveness of the event-triggered control design is shown.

The following summary shows the highlights of this design:

- (i) By considering the NCSs with missing data and transmission delay, the problem of the IT2 fuzzy event-triggered stabilization is investigated.
- (ii) Compared with the existing works on fuzzy control, the plant is subject to parameter uncertainties. Compared with the existing works on IT2 fuzzy control, the event-triggered controller is under missing data and transmission delay. Thus, the controlled systems and the controller design are both more complex.
- (iii) In contrast to the presented IT2 fuzzy sampled-data controllers for NCSs in [11–16], the designed IT2 fuzzy event-triggered controller can further save the implementation cost with the reduced sampling number and limited bandwidth.

Notations: The matrix $N > 0$ denotes that N is positive, N^T is the transpose of N , the inverse of N is expressed by N^{-1} , and $*$ stands for a symmetric element.

2. Problem formulation

The IT2 fuzzy model converts the NCSs into some linear models with the form (1):

Rule i : IF $h_1(m(t))$ is $\hat{H}_1^i$ and,..., and $h_\lambda(m(t))$ is $\hat{H}_\lambda^i$, THEN

$$\dot{x}(t) = A_i x(t) + B_i u(t), i = 1, 2, \dots, r, \quad (1)$$

where $x(t)$ denotes the state, $u(t)$ means the input, A_i and B_i are time-invariant matrices, $h_1(m(t)), \dots, h_\lambda(m(t))$ are premise variables, and $\hat{H}_1^i, \dots, \hat{H}_\lambda^i$ are fuzzy sets.

The global system of NCSs is expressed as

$$\dot{x}(t) = \sum_{i=1}^r \vartheta_i(m(t)) [A_i x(t) + B_i u(t)], \quad (2)$$

where

$$\vartheta_i(m(t)) = \vartheta_i^l(m(t)) \underline{c}_i(\varepsilon(t)) + \vartheta_i^u(m(t)) \bar{c}_i(m(t)), \quad (3)$$

$$\vartheta_i^l(m(t)) = \underline{\theta}_{\hat{H}_1^i}(h_1(m(t))) \times \underline{\theta}_{\hat{H}_2^i}(h_2(m(t))) \times \dots \times \underline{\theta}_{\hat{H}_\lambda^i}(h_\lambda(m(t))) \geq 0, \quad (4)$$

$$\vartheta_i^u(m(t)) = \bar{\theta}_{\hat{H}_1^i}(h_1(m(t))) \times \bar{\theta}_{\hat{H}_2^i}(h_2(m(t))) \times \dots \times \bar{\theta}_{\hat{H}_\lambda^i}(h_\lambda(m(t))) \geq 0, \quad (5)$$

$\sum_{i=1}^r \vartheta_i(m(t)) = 1$, $\underline{c}_i(m(t)) + \bar{c}_i(m(t)) = 1$, $0 \leq \underline{c}_i(m(t)) \leq 1$, $0 \leq \bar{c}_i(m(t)) \leq 1$, and the membership functions $0 \leq \underline{\theta}_{\hat{H}_l^i}(h_l(m(t))) \leq 1$ and $0 \leq \bar{\theta}_{\hat{H}_l^i}(h_l(m(t))) \leq 1$, $l = 1, 2, \dots, \lambda$.

The sensors in NCSs are time-triggered, where the sampling period is a constant h . The sampling instants are $0, h, 2h, \dots$. An event-triggered transmission mechanism is

$$t_{k+1}h = t_kh + \min \{lh | e^T(i_kh) \Psi e(i_kh) \geq \delta x^T(t_kh) \Psi x(t_kh)\}, \quad (6)$$

where $t_{k+1} \in \mathbb{N}$, $t_k \in \mathbb{N}$, and $i_k \in \mathbb{N}$; $e(i_kh) = x(i_kh) - x(t_kh)$ is an error with $i_k = t_k + l$, $l \in \mathbb{N}$, $x(i_kh)$ denotes the current sampling state, and $x(t_kh)$ expresses the latest transmission state; and the matrix $\Psi > 0$ and the parameter $\delta \in [0, 1)$.

Now, design an IT2 fuzzy event-triggered controller for NCSs for the fuzzy system (2). The j th rule is defined as

Rule j : IF $h_1(m(t_kh))$ is $\hat{H}_1^j$ and,..., and $h_\lambda(m(t_kh))$ is $\hat{H}_\lambda^j$, THEN

$$u(t) = K_j x(t_kh), t \in [t_kh + \tau_{t_k}, t_{k+1}h + \tau_{t_{k+1}}) \quad (7)$$

where K_j is the feedback gain; $t_k h$, $k = 0, 1, 2, \dots$ denote transmission instants, the phenomena of data dropout occurs from $t_k h$ to $t_{k+1} h$, and the data dropout rate is σ ; τ_k is the transmission delay and $0 < \xi \leq \tau_k \leq \bar{\tau}$; and $j = 1, 2, \dots, r$.

One has the overall controller

$$u(t) = \sum_{j=1}^r (\vartheta_j(m(t_k h)) + \bar{\vartheta}_j(m(t_k h))) K_j x(t_k h), t \in [t_k h + \tau_{t_k}, t_{k+1} h + \tau_{t_{k+1}}) \quad (8)$$

in which

$$\vartheta_j(m(t_k h)) = \frac{(\underline{\rho}_j(m(t_k h)) \vartheta_j^l(m(t_k h)))}{\sum_{j=1}^r \underline{\rho}_j(m(t_k h)) \vartheta_j^l(m(t_k h)) + \bar{\rho}_j(m(t_k h)) \vartheta_j^u(m(t_k h))},$$

$$\bar{\vartheta}_j(m(t_k h)) = \frac{\bar{\rho}_j(m(t_k h)) \vartheta_j^u(m(t_k h))}{\sum_{j=1}^r (\underline{\rho}_j(m(t_k h)) \vartheta_j^l(m(t_k h)) + \bar{\rho}_j(m(t_k h)) \vartheta_j^u(m(t_k h)))}.$$

For a detailed analysis, $[t_k h + \tau_{t_k}, t_{k+1} h + \tau_{t_{k+1}})$ is divided into some subsets Ω_l with the form

$$\Omega_l = [i_k h + \tau_{i_k}, (i_k + 1)h + \tau_{i_{k+1}}),$$

where $i_k = t_k + l$, $l = 0, 1, \dots, t_{k+1} - t_k - 1$; if $l = t_{k+1} - t_k - 1$, then $\tau_{i_{k+1}} = \tau_{t_{k+1}}$, otherwise, $\tau_{i_k} = \tau_{t_k}$; and $[t_k h + \tau_{t_k}, t_{k+1} h + \tau_{t_{k+1}}) = \bigcup_{l=0}^{t_{k+1}-t_k-1} \Omega_l$.

Let $\xi(t) = t - i_k h$, $t \in \Omega_l$, one has the controller with the form (9):

$$u(t) = \sum_{j=1}^r (\vartheta_j(m(t_k h)) + \bar{\vartheta}_j(m(t_k h))) K_j (x(t - \xi(t)) - e(i_k h)), \quad (9)$$

in which $0 < \underline{\xi} \leq \xi(t) \leq h + \bar{\tau} = \bar{\xi}$, and $\dot{\xi}(t) = 1$.

Using (2) and (9), we have the closed-loop system

$$\dot{x}(t) = \sum_{i=1}^r \sum_{j=1}^r \vartheta_i(m(t)) (\vartheta_j(m(t_k h)) + \bar{\vartheta}_j(m(t_k h))) [A_i x(t) + B_i K_j x(t - \xi(t)) - B_i K_j e(i_k h)]. \quad (10)$$

Lemma 1 [25]. For any matrices F_1, F_2, F_3 , any $s \in [0, 1]$, then $F_1 + sF_2 + (1-s)F_3 < 0$ holds if and only if $F_1 + F_2 < 0$ and $F_1 + F_3 < 0$ hold.

3. IT2 fuzzy event-triggered stabilization

This part investigates the stabilization problem for NCSs under the event-triggered mechanism and a fuzzy controller will be presented. $\vartheta_i(m(t))$, $\vartheta_j(m(t_k h))$, and $\bar{\vartheta}_j(m(t_k h))$ are denoted by ϑ_i , ϑ_j , and $\bar{\vartheta}_j$, correspondingly.

Theorem 1. For scalars $h > 0$, $\delta > 0$, $\underline{\xi} > 0$, $\bar{\xi} > 0$, $\sigma > 0$, and α , the control system (10) is of asymptotic stability if there are scalars $\gamma_i > 0$ subjected to $\gamma_i \leq (\vartheta_i + \bar{\vartheta}_i)/\vartheta_i$, matrices $\bar{X} > 0$, $\bar{N} > 0$, $\bar{P} = \begin{bmatrix} \bar{P}_1 & \bar{P}_2 \\ * & \bar{P}_3 \end{bmatrix} \geq 0$, $\bar{S}_1 > 0$, $\bar{S}_2 > 0$, $\bar{W}_1 > 0$, $\bar{W}_2 > 0$, $\bar{\Psi} > 0$, and

appropriately matrices $\bar{U}$, $\bar{T}_i = [\bar{T}_{i1}^T \quad \bar{T}_{i2}^T \quad \bar{T}_{i3}^T \quad 0 \quad 0 \quad 0]^T$, $\bar{\Pi}_i = \bar{\Pi}_i^T$, and $\bar{K}_j$ satisfy the conditions (11)–(14) for all $1 \leq i, j \leq r$:

$$\gamma_j \bar{\Gamma}_{ij} - \gamma_j \bar{\Pi}_i + \bar{\Pi}_i + \bar{\xi} \gamma_j \bar{\Delta} < 0, \quad (11)$$

$$\begin{bmatrix} \gamma_j \bar{\Gamma}_{ij} - \gamma_j \bar{\Pi}_i + \bar{\Pi}_i & \bar{\xi} \sqrt{\gamma_j} \bar{T}_i \\ * & -\bar{\xi} \bar{N} \end{bmatrix} < 0, \quad (12)$$

$$\bar{\Gamma}_{ij} - \bar{\Pi}_i + \bar{\xi} \bar{\Delta} < 0, \quad (13)$$

$$\begin{bmatrix} \bar{\Gamma}_{ij} - \bar{\Pi}_i & \bar{\xi} \bar{T}_i \\ * & -\bar{\xi} \bar{N} \end{bmatrix} < 0, \quad (14)$$

where

$$\bar{\Gamma}_{ij} = \begin{bmatrix} \bar{\Gamma}_{ij11} & \bar{\Gamma}_{ij12} & \bar{\Gamma}_{ij13} & \bar{\Gamma}_{ij14} & 0 & 0 \\ * & \bar{\Gamma}_{ij22} & \bar{\Gamma}_{ij23} & \bar{\Gamma}_{ij24} & \bar{\Gamma}_{ij25} & \bar{\Gamma}_{ij26} \\ * & * & \bar{\Gamma}_{ij33} & 0 & 0 & \bar{\Gamma}_{ij36} \\ * & * & * & \bar{\Gamma}_{ij44} & 0 & 0 \\ * & * & * & * & \bar{\Gamma}_{ij55} & 0 \\ * & * & * & * & * & \bar{\Gamma}_{ij66} \end{bmatrix}, \quad \bar{\Delta} = \begin{bmatrix} 0 & 0 & \bar{P}_1 & 0 & 0 & 0 \\ * & 0 & \bar{P}_2^T & 0 & 0 & 0 \\ * & * & \bar{N} & 0 & 0 & 0 \\ * & * & * & 0 & 0 & 0 \\ * & * & * & * & 0 & 0 \\ * & * & * & * & * & 0 \end{bmatrix}$$

with

$$\begin{aligned} \bar{\Gamma}_{ij11} &= -\bar{P}_1 + \bar{T}_{i1} + \bar{T}_{i1}^T + \bar{S}_1 + \bar{S}_2 - \bar{W}_1, & \bar{\Gamma}_{ij12} &= -\bar{T}_{i1} + \bar{T}_{i2}^T - \bar{P}_2 + \alpha \bar{U} A_i^T, & \bar{\Gamma}_{ij13} &= \bar{U} A_i^T + \bar{X} + \bar{T}_{i3}^T, \\ \bar{\Gamma}_{ij14} &= \bar{W}_1, & \bar{\Gamma}_{ij22} &= -\bar{P}_3 - \bar{T}_{i2} - \bar{T}_{i2}^T + \alpha B_i \bar{K}_j + \alpha \bar{K}_j^T B_i^T - 2\bar{W}_1 - 2\bar{W}_2 + \delta \bar{\Psi}, \\ \bar{\Gamma}_{ij23} &= -\alpha \bar{U} - \bar{T}_{i3}^T + \bar{K}_j^T B_i^T, & \bar{\Gamma}_{ij24} &= \bar{W}_1 + \bar{W}_2, & \bar{\Gamma}_{ij25} &= \bar{W}_2, & \bar{\Gamma}_{ij26} &= -\delta \bar{\Psi} - \alpha B_i \bar{K}_j, \\ \bar{\Gamma}_{ij33} &= -\bar{U} - \bar{U}^T + \bar{\xi}^2 \bar{W}_1 + (\bar{\xi} - \xi)^2 \bar{W}_2, & \bar{\Gamma}_{ij36} &= -B_i \bar{K}_j, & \bar{\Gamma}_{ij44} &= -\bar{S}_1 - \bar{W}_1 - \bar{W}_2, \\ \bar{\Gamma}_{ij55} &= -\bar{S}_2 - \bar{W}_2, \\ \bar{\Gamma}_{ij66} &= -(1 - \delta) \bar{\Psi}. \end{aligned}$$

There is $K_j = \bar{K}_j (\bar{U}^T)^{-1}$.

Proof. Construct the LKF

$$V(x) = \sum_{s=1}^4 V_s(x), \quad (15)$$

where

$$V_1(x) = x^T(t) X x(t), \quad (16)$$

$$V_2(x) = (\bar{\xi} - \xi(t)) \int_{t-\xi(t)}^t x^T(s) N \dot{x}(s) ds, \quad (17)$$

$$V_3(x) = (\bar{\xi} - \xi(t)) \hat{x}^T(t) P \hat{x}(t), \quad (18)$$

$$V_4(x) = \int_{t-\underline{\xi}}^t x^T(s) S_1 x(s) ds + \int_{t-\bar{\xi}}^t x^T(s) S_2 x(s) ds + \underline{\xi} \int_{-\underline{\xi}}^0 \int_{t+\theta}^t \dot{x}^T(s) W_1 \dot{x}(s) ds d\theta \\ + (\bar{\xi} - \underline{\xi}) \int_{-\bar{\xi}}^{-\underline{\xi}} \int_{t+\theta}^t \dot{x}^T(s) W_2 \dot{x}(s) ds d\theta \quad (19)$$

with $X > 0$, $N > 0$, $P = \begin{bmatrix} P_1 & P_2 \\ * & P_3 \end{bmatrix} \geq 0$, $S_1 > 0$, $S_2 > 0$, $W_1 > 0$, $W_2 > 0$, $\hat{x}(t) = [x^T(t) \quad x_k^T(t)]^T$, and $x_k^T(t) = x(t_k)$.

Obviously, $V(x) > 0$.

Noting that $x_k(t) = x(t_k h)$, one has $\dot{x}_k(t) = 0$ with the constant $x(t_k h)$. More, $\dot{\xi}(t) = 1$ for $t \in (t_k h + \tau_{t_k}, t_{k+1} h + \tau_{t_{k+1}})$. There are

$$\dot{V}_1(x) = 2x^T(t) X \dot{x}(t), \quad (20)$$

$$\dot{V}_2(x) = (\bar{\xi} - \xi(t)) \dot{x}^T(t) N \dot{x}(t) - \int_{t-\xi(t)}^t \dot{x}^T(s) N \dot{x}(s) ds, \quad (21)$$

$$\dot{V}_3(x) = (\bar{\xi} - \xi(t)) 2\hat{x}^T(t) P \hat{x}(t) - \hat{x}^T(t) P \hat{x}(t), \quad (22)$$

$$\dot{V}_4(x) = x^T(t) (S_1 + S_2) x(t) - x^T(t - \underline{\xi}) S_1 x(t - \underline{\xi}) - x^T(t - \bar{\xi}) S_2 x(t - \bar{\xi}) + \dot{x}(t)^T (\underline{\xi}^2 W_1 \\ + (\bar{\xi} - \underline{\xi})^2 W_2) \dot{x}(t) - \underline{\xi} \int_{t-\underline{\xi}}^t \dot{x}^T(s) W_1 \dot{x}(s) ds - (\bar{\xi} - \underline{\xi}) \int_{t-\bar{\xi}}^{t-\underline{\xi}} \dot{x}^T(s) W_2 \dot{x}(s) ds \\ \leq x^T(t) (S_1 + S_2) x(t) - x^T(t - \underline{\xi}) S_1 x(t - \underline{\xi}) - x^T(t - \bar{\xi}) S_2 x(t - \bar{\xi}) \\ + \dot{x}(t)^T (\underline{\xi}^2 W_1 + (\bar{\xi} - \underline{\xi})^2 W_2) \dot{x}(t) \\ - (x(t) - x(t - \xi(t)))^T W_1 (x(t) - x(t - \xi(t))) - (x(t - \xi(t)))^T W_1 (x(t - \xi(t)) - x(t - \underline{\xi})) \\ - (x(t - \underline{\xi}))^T W_1 (x(t - \xi(t)) - x(t - \underline{\xi})) - (x(t - \underline{\xi}))^T W_2 (x(t - \xi(t)) - x(t - \xi(t))) \\ - (x(t - \xi(t)))^T W_2 (x(t - \xi(t)) - x(t - \bar{\xi})). \quad (23)$$

Let $\tilde{x}(t) = [x^T(t) \quad x^T(t - \xi(t)) \quad \dot{x}^T(t)]^T$. For any appropriately dimensioned matrices $\underline{T}_i = [T_{i1}^T \quad T_{i2}^T \quad T_{i3}^T]^T$, $i = 1, 2, \dots, r$, we have

$$\int_{t-\xi(t)}^t [\tilde{x}(t) \quad \dot{x}(s)] \begin{bmatrix} \underline{T}_i N^{-1} \underline{T}_i^T & \underline{T}_i \\ * & N \end{bmatrix} \begin{bmatrix} \tilde{x}^T(t) & \dot{x}^T(s) \end{bmatrix}^T ds \geq 0. \quad (24)$$

Thus,

$$- \int_{t-\xi(t)}^t \dot{x}^T(s) N \dot{x}(s) ds \leq \xi(t) \tilde{x}^T(t) \underline{T}_i N^{-1} \underline{T}_i^T \tilde{x}(t) + 2\tilde{x}(t) \underline{T}_i [x(t) - x(t - \xi(t))]. \quad (25)$$

A zero item holds

$$0 = 2\tilde{x}^T(t) \tilde{U} \left\{ \sum_{i=1}^r \sum_{j=1}^r \vartheta_i (\vartheta_j + \bar{\vartheta}_j) [A_i x(t) + B_i K_j x(t - \xi(t)) - B_i K_j e(i_k h)] - \dot{x}(t) \right\}, \quad (26)$$

in which $\tilde{U} = [0 \quad \alpha U^T \quad U^T]^T$, α is a scalar, and the matrix U is nonsingular.

For the event-triggered mechanism, there is

$$0 \leq \delta(x(t - \xi(t)) - e(i_k h))^T \Psi(x(t - \xi(t)) - e(i_k h)) - e^T(i_k h) \Psi e(i_k h). \quad (27)$$

By use of (20)–(27), one obtains

$$\dot{V} \leq \sum_{i=1}^r \sum_{j=1}^r \vartheta_i(\vartheta_j + \bar{\vartheta}_j) \bar{x}^T(t) \left[\tilde{\Gamma}_{ij} + \xi(t) T_i N^{-1} T_i^T + (\bar{\xi} - \xi(t)) \Delta \right] \bar{x}(t), \quad (28)$$

where

$$\begin{aligned} \bar{x}(t) &= [x^T(t) \quad x^T(t - \xi(t)) \quad \bar{x}^T(t) \quad x^T(t - \bar{\xi}) \quad x^T(t - \bar{\xi}) \quad e(i_k h)]^T, \\ T_i &= [T_{i1}^T \quad T_{i2}^T \quad T_{i3}^T \quad 0 \quad 0 \quad 0]^T, \quad i = 1, 2, \dots, r \\ \tilde{\Gamma}_{ij} &= \begin{bmatrix} \tilde{\Gamma}_{ij11} & \tilde{\Gamma}_{ij12} & \tilde{\Gamma}_{ij13} & \tilde{\Gamma}_{ij14} & 0 & 0 \\ * & \tilde{\Gamma}_{ij22} & \tilde{\Gamma}_{ij23} & \tilde{\Gamma}_{ij24} & \tilde{\Gamma}_{ij25} & \tilde{\Gamma}_{ij26} \\ * & * & \tilde{\Gamma}_{ij33} & 0 & 0 & \tilde{\Gamma}_{ij36} \\ * & * & * & \tilde{\Gamma}_{ij44} & 0 & 0 \\ * & * & * & * & \tilde{\Gamma}_{ij55} & 0 \\ * & * & * & * & * & \tilde{\Gamma}_{ij66} \end{bmatrix}, \quad \Delta = \begin{bmatrix} 0 & 0 & P_1 & 0 & 0 & 0 \\ * & 0 & P_2^T & 0 & 0 & 0 \\ * & * & \bar{N} & 0 & 0 & 0 \\ * & * & * & 0 & 0 & 0 \\ * & * & * & * & 0 & 0 \\ * & * & * & * & * & 0 \end{bmatrix} \end{aligned}$$

with

$$\begin{aligned} \tilde{\Gamma}_{ij11} &= -P_1 + T_{i1} + T_{i1}^T + S_1 + S_2 - W_1, \quad \tilde{\Gamma}_{ij12} = -T_{i1} + T_{i2}^T - P_2 + \alpha A_i^T U^T, \quad \tilde{\Gamma}_{ij13} = A_i^T U^T + X + T_{i3}^T, \\ \tilde{\Gamma}_{ij14} &= W_1, \quad \tilde{\Gamma}_{ij22} = -P_3 - T_{i2} - T_{i2}^T + \alpha U B_i K_j + \alpha K_j^T B_i^T U^T - 2W_1 - 2W_2 + \delta \Psi, \\ \tilde{\Gamma}_{ij23} &= -\alpha U - T_{i3}^T + K_j^T B_i^T U^T, \quad \tilde{\Gamma}_{ij24} = W_1 + W_2, \quad \tilde{\Gamma}_{ij25} = W_2, \quad \tilde{\Gamma}_{ij26} = -\delta \Psi - \alpha U B_i K_j, \\ \tilde{\Gamma}_{ij33} &= -U - U^T + \bar{\xi}^2 W_1 + (\bar{\xi} - \xi)^2 W_2, \quad \tilde{\Gamma}_{ij36} = -U B_i K_j, \quad \tilde{\Gamma}_{ij44} = -S_1 - W_1 - W_2, \\ \tilde{\Gamma}_{ij55} &= -S_2 - W_2, \\ \tilde{\Gamma}_{ij66} &= -(1 - \delta) \Phi. \end{aligned}$$

And then, the membership functions are applied to improve this stability design.

Let

$$\Upsilon_0 = \sum_{i=1}^r \sum_{j=1}^r \vartheta_i(\vartheta_j + \bar{\vartheta}_j) \left[\tilde{\Gamma}_{ij} + \xi(t) T_i N^{-1} T_i^T + (\bar{\xi} - \xi(t)) \Delta \right]. \quad (29)$$

Noting that $\sum_{i=1}^r \vartheta_i = \sum_{j=1}^r (\vartheta_j + \bar{\vartheta}_j) = 1$, one has $\sum_{i=1}^r (\vartheta_i - (\vartheta_i + \bar{\vartheta}_i)) = 0$. Thus, the following equality holds:

$$\Upsilon_1 = \sum_{i=1}^r \sum_{j=1}^r \vartheta_i(\vartheta_j - (\vartheta_j + \bar{\vartheta}_j)) \bar{\Pi}_i = 0. \quad (30)$$

where $\bar{\Pi}_i, i = 1, 2, \dots, r$ are symmetrical slack matrices.

Using (29)–(30), there is

$$\begin{aligned} \Upsilon_0 &= \Upsilon_0 + \Upsilon_1 \\ &= \sum_{i=1}^r \sum_{j=1}^r \vartheta_i(\vartheta_j + \bar{\vartheta}_j) \left[\tilde{\Gamma}_{ij} + \xi(t) T_i N^{-1} T_i^T + (\bar{\xi} - \xi(t)) \Delta \right] + \sum_{i=1}^r \sum_{j=1}^r \vartheta_i(\vartheta_j - (\vartheta_j + \bar{\vartheta}_j)) \Pi_i \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^r \sum_{j=1}^r \vartheta_i((\underline{\vartheta}_j + \bar{\vartheta}_j) - \gamma_j \vartheta_j + \gamma_j \vartheta_j) \left[\tilde{\Gamma}_{ij} + \xi(t) T_i N^{-1} T_i^T + (\bar{\xi} - \xi(t)) \Delta \right] \\
&\quad + \sum_{i=1}^r \sum_{j=1}^r \vartheta_i(\vartheta_j - (\underline{\vartheta}_j + \bar{\vartheta}_j) - \gamma_j \vartheta_j + \gamma_j \vartheta_j) \Pi_i \\
&= \sum_{i=1}^r \sum_{j=1}^r \vartheta_i \vartheta_j \left\{ \gamma_j \left[\tilde{\Gamma}_{ij} + \xi(t) T_i N^{-1} T_i^T + (\bar{\xi} - \xi(t)) \Delta \right] - \gamma_j \Pi_i + \Pi_i \right\} \\
&\quad + \sum_{i=1}^r \sum_{j=1}^r \vartheta_i((\underline{\vartheta}_j + \bar{\vartheta}_j) - \gamma_j \vartheta_j) \left[\tilde{\Gamma}_{ij} + \xi(t) T_i N^{-1} T_i^T + (\bar{\xi} - \xi(t)) \Delta - \Pi_i \right] \\
&= \sum_{i=1}^r \sum_{j=1}^r \vartheta_i \vartheta_j \Omega_{1ij} + \sum_{i=1}^r \sum_{j=1}^r \vartheta_i((\underline{\vartheta}_j + \bar{\vartheta}_j) - \gamma_j \vartheta_j) \Omega_{2ij}, \tag{31}
\end{aligned}$$

where

$$\Omega_{1ij} = \gamma_j \left[\tilde{\Gamma}_{ij} + \xi(t) T_i N^{-1} T_i^T + (\bar{\xi} - \xi(t)) \Delta \right] - \gamma_j \Pi_i + \Pi_i,$$

$$\Omega_{2ij} = \left[\tilde{\Gamma}_{ij} + \xi(t) T_i N^{-1} T_i^T + (\bar{\xi} - \xi(t)) \Delta - \Pi_i \right].$$

With Lemma 1 in [25], $\Omega_{1ij} < 0$ is converted into

$$\gamma_j \tilde{\Gamma}_{ij} - \gamma_j \Pi_i + \Pi_i + \bar{\xi} \gamma_j \Delta < 0, \tag{32}$$

$$\begin{bmatrix} \gamma_j \tilde{\Gamma}_{ij} - \gamma_j \Pi_i + \Pi_i & \bar{\xi} \sqrt{\gamma_j} T_i \\ * & -\bar{\xi} N \end{bmatrix} < 0. \tag{33}$$

Using Lemma 1 in [25], $\Omega_{2ij} < 0$ is equivalent to

$$\tilde{\Gamma}_{ij} - \Pi_i + \bar{\xi} \Delta < 0, \tag{34}$$

$$\begin{bmatrix} \tilde{\Gamma}_{ij} - \Pi_i & \bar{\xi} T_i \\ * & -\bar{\xi} N \end{bmatrix} < 0. \tag{35}$$

Let

$$\bar{U} = U^{-1}, \quad \bar{X} = \bar{U} X \bar{U}^T, \quad \bar{N} = \bar{U} N \bar{U}^T, \quad \bar{P}_1 = \bar{U} P_1 \bar{U}^T, \quad \bar{P}_2 = \bar{U} P_2 \bar{U}^T, \quad \bar{P}_3 = \bar{U} P_3 \bar{U}^T, \quad \bar{S}_1 = \bar{U} S_1 \bar{U}^T, \quad \bar{S}_2 = \bar{U} S_2 \bar{U}^T,$$

$$\bar{W}_1 = \bar{U} W_1 \bar{U}^T, \quad \bar{W}_2 = \bar{U} W_2 \bar{U}^T, \quad \bar{T}_{11} = \bar{U} T_{11} \bar{U}^T, \quad \bar{T}_{12} = \bar{U} T_{12} \bar{U}^T, \quad \bar{T}_{13} = \bar{U} T_{13} \bar{U}^T, \quad \bar{\Psi} = \bar{U} \Psi \bar{U}^T, \quad \text{and } \bar{K}_j = \bar{K}_j \bar{U}^T.$$

Pre- and post-multiplying $\tilde{\Gamma}_{ij}$, Π_i , and Δ by $\text{diag}[\bar{U} \quad \bar{U} \quad \bar{U} \quad \bar{U} \quad \bar{U} \quad \bar{U}]$ and $\text{diag}[\bar{U}^T \quad \bar{U}^T \quad \bar{U}^T \quad \bar{U}^T \quad \bar{U}^T \quad \bar{U}^T]$ result in $\bar{\Gamma}_{ij}$, $\bar{\Pi}_i$, and $\bar{\Delta}$, respectively. Obviously, (32) and (34) are equivalent to (11) and (13), respectively.

Pre- and post-multiplying (33) and (35) by $\text{diag}[\bar{U} \quad \bar{U} \quad \bar{U} \quad \bar{U} \quad \bar{U} \quad \bar{U}]$ and $\text{diag}[\bar{U}^T \quad \bar{U}^T \quad \bar{U}^T \quad \bar{U}^T \quad \bar{U}^T \quad \bar{U}^T]$ and using the Schur Complement, (33) and (35) are equivalent to (12) and (14), respectively.

Thus, there are $\Omega_{1ij} < 0$ and $\Omega_{2ij} < 0$. With $\Omega_{1ij} < 0$, $\Omega_{2ij} < 0$, and $\gamma_i \leq (\vartheta_i + \bar{\vartheta}_i)/\vartheta_i$, one has $\Upsilon_0 < 0$. As a result, $\dot{V} < 0$ for $t \in (t_k + \tau_k, t_{k+1} + \tau_{k+1})$.

Owing to the continuity of the LKF V , it can be concluded that $\dot{V} < 0$ for $t \in [t_0, \infty)$. $\square$

Remark 1. Once the feedback gains are determined, the control design of this paper is effective for the different data packet loss rates. However, in the paper [25], if the data packet loss rate varies, then the feedback gains need to be solved again. In contrast, this paper shows its superiority.

Remark 2. If we consider the data dropout rate $\sigma = 0$ and the time delay $\tau_k = 0$, $k = 0, 1, 2, \dots$, the system (10) becomes a sampled-data control system. As for this system, there is [Corollary 1](#).

Remark 3. In this design, the sensors in NCSs are time-triggered, where the sampling period is a constant h . That is to say, the signal is sampled in a fixed period in the networked environment. The lower bound of the event-triggered mechanism is h , and the signals are not triggered infinite times in a very short time.

Remark 4. In this control system design, the LKF is constructed to carry out system stability, and the free-weighting matrix and some slack matrices are introduced to improve stability analysis. The obtained results will be less conservative. The dimension of LMIs of this paper is low, and the number of the parameters on membership functions is small. Thus, there is not the problem of computational complexity in this design.

Corollary 1. For scalars $h > 0$, $\delta > 0$, and α , the system (10) is of asymptotic stability if there exist scalars $\gamma_i > 0$ subjected to $\gamma_i \leq (\vartheta_i + \bar{\vartheta}_i)/\vartheta_i$, matrices $\bar{X} > 0$, $\bar{N} > 0$, $\bar{P} = \begin{bmatrix} \bar{P}_1 & \bar{P}_2 \\ * & \bar{P}_3 \end{bmatrix} \geq 0$, $\bar{\Psi} > 0$, and appropriately matrices $\bar{U}$, $\hat{T}_i = [\bar{T}_{i1}^T \quad \bar{T}_{i2}^T \quad \bar{T}_{i3}^T \quad 0]^T$, $\hat{\Pi}_i = \hat{\Pi}_i^T$, and $\bar{K}_j$ such that for all $1 \leq i, j \leq r$, the conditions (36)–(39) are satisfied,

$$\gamma_j \hat{\Gamma}_{ij} - \gamma_j \hat{\Pi}_i + \hat{\Pi}_i + h\gamma_j \hat{\Delta} < 0, \quad (36)$$

$$\begin{bmatrix} \gamma_j \hat{\Gamma}_{ij} - \gamma_j \hat{\Pi}_i + \hat{\Pi}_i & h\sqrt{\gamma_j} \hat{T}_i \\ * & -h\bar{N} \end{bmatrix} < 0, \quad (37)$$

$$\hat{\Gamma}_{ij} - \hat{\Pi}_i + h\hat{\Delta} < 0, \quad (38)$$

$$\begin{bmatrix} \hat{\Gamma}_{ij} - \hat{\Pi}_i & h\hat{T}_i \\ * & -h\bar{N} \end{bmatrix} < 0, \quad (39)$$

where

$$\hat{\Gamma}_{ij} = \begin{bmatrix} \hat{\Gamma}_{ij11} & \hat{\Gamma}_{ij12} & \hat{\Gamma}_{ij13} & 0 \\ * & \hat{\Gamma}_{ij22} & \hat{\Gamma}_{ij23} & \hat{\Gamma}_{ij24} \\ * & * & \hat{\Gamma}_{ij33} & \hat{\Gamma}_{ij34} \\ * & * & * & \hat{\Gamma}_{ij44} \end{bmatrix}, \quad \bar{\Delta} = \begin{bmatrix} 0 & 0 & \bar{P}_1 & 0 \\ * & 0 & \bar{P}_2^T & 0 \\ * & * & \bar{N} & 0 \\ * & * & * & 0 \end{bmatrix}$$

with

$$\begin{aligned} \hat{\Gamma}_{ij11} &= -\bar{P}_1 + \bar{T}_{i1} + \bar{T}_{i1}^T, \quad \hat{\Gamma}_{ij12} = -\bar{T}_{i1} + \bar{T}_{i2}^T - \bar{P}_2 + \alpha \bar{U} A_i^T, \quad \hat{\Gamma}_{ij13} = \bar{U} A_i^T + \bar{X} + \bar{T}_{i3}^T, \\ \hat{\Gamma}_{ij22} &= -\bar{P}_3 - \bar{T}_{i2} - \bar{T}_{i2}^T + \alpha B_i \bar{K}_j + \alpha \bar{K}_j^T B_i^T + \delta \bar{\Psi}, \quad \hat{\Gamma}_{ij23} = -\alpha \bar{U} - \bar{T}_{i3}^T + \bar{K}_j^T B_i^T, \quad \hat{\Gamma}_{ij24} = \\ &= -\delta \bar{\Psi} - \alpha B_i \bar{K}_j, \end{aligned}$$

$$\hat{\Gamma}_{ij33} = -\bar{U} - \bar{U}^T, \hat{\Gamma}_{ij34} = -B_i \bar{K}_j, \hat{\Gamma}_{ij44} = -(1 - \delta) \bar{\Psi}.$$

One has $K_j = \bar{K}_j (\bar{U}^T)^{-1}$.

4. Simulation examples

The following experiments show the feasibility and the superiority of this design.

Example 1. Consider the continuous stirred tank reactor (CSTR)

$$\begin{aligned} \dot{z}_1 &= -z_1 + \bar{D}_a(1 - z_1)e^{-\frac{1}{z_2 + \bar{\gamma}}}, \\ \dot{z}_2 &= -(1 + \bar{\beta})z_2 + \bar{H}\bar{D}_a(1 - z_1)e^{-\frac{1}{z_2 + \bar{\gamma}}} + \bar{\beta}\bar{u}, \end{aligned} \quad (40)$$

where $z = [z_1 \ z_2]^T$, $\bar{u}$ is associated with the inputs, and the parameters $D_a = 0.072$, $\bar{H} = 8$, $\bar{\gamma} = 20$ and $\bar{\beta} = 0.3$.

Let $x = [x_1 \ x_2]^T = z - z_e$ and $u = \bar{u} - u_e$. The operating point is $z_e = (0.1440, 0.8862)$ with $u_e = 0$. The system is modelled as the fuzzy system as follows:

Rule 1: IF the temperature is low (i.e., z_2 is about -1), THEN,

$$\dot{x}(t) = A_1 x(t) + B_1 u(t),$$

Rule 2: IF the temperature is middle (i.e., z_2 is about 0), THEN,

$$\dot{x}(t) = A_2 x(t) + B_2 u(t),$$

Rule 3: IF the temperature is high (i.e., z_2 is about 1), THEN,

$$\dot{x}(t) = A_3 x(t) + B_3 u(t),$$

where

$$\begin{aligned} A_1 &= \begin{bmatrix} -1.0684 & 0.0002 \\ -0.5471 & 1.2987 \end{bmatrix}, B_1 = \begin{bmatrix} 0 \\ 0.3 \end{bmatrix}, A_2 = \begin{bmatrix} -1.0686 & 0.0002 \\ -0.5484 & -1.2988 \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ 0.3 \end{bmatrix}, \\ A_3 &= \begin{bmatrix} -1.0687 & 0.0002 \\ -0.5497 & 1.2988 \end{bmatrix}, B_3 = \begin{bmatrix} 0 \\ 0.3 \end{bmatrix}. \end{aligned}$$

The membership functions are shown as

$$\begin{aligned} \vartheta_1^l(z_2) &= \frac{e^{-\frac{(z_2+1)^2}{0.5^2}}}{e^{-\frac{(z_2+1)^2}{0.5^2}} + e^{-\frac{z_2^2}{0.8^2}} + e^{-\frac{(z_2-1)^2}{0.5^2}}}, \vartheta_2^l(z_2) = \frac{e^{-\frac{z_2^2}{0.8^2}}}{e^{-\frac{(z_2+1)^2}{0.5^2}} + e^{-\frac{z_2^2}{0.8^2}} + e^{-\frac{(z_2-1)^2}{0.5^2}}}, \\ \vartheta_3^l(z_2) &= \frac{e^{-\frac{(z_2-1)^2}{0.5^2}}}{e^{-\frac{(z_2+1)^2}{0.5^2}} + e^{-\frac{z_2^2}{0.5^2}} + e^{-\frac{(z_2-1)^2}{0.5^2}}}, \vartheta_1^u(z_2) = \frac{e^{-\frac{(z_2+1)^2}{0.8^2}}}{e^{-\frac{(z_2+1)^2}{0.8^2}} + e^{-\frac{z_2^2}{0.8^2}} + e^{-\frac{(z_2-1)^2}{0.8^2}}}, \\ \vartheta_2^u(z_2) &= \frac{e^{-\frac{z_2^2}{0.5^2}}}{e^{-\frac{(z_2+1)^2}{0.5^2}} + e^{-\frac{z_2^2}{0.5^2}} + e^{-\frac{(z_2-1)^2}{0.5^2}}}, \vartheta_3^u(z_2) = \frac{e^{-\frac{(z_2-1)^2}{0.8^2}}}{e^{-\frac{(z_2+1)^2}{0.8^2}} + e^{-\frac{z_2^2}{0.8^2}} + e^{-\frac{(z_2-1)^2}{0.8^2}}}, \end{aligned}$$

with $\underline{c}_i(t) = \sin^2(z_2)$, $\bar{c}_i(t) = 1 - \sin^2(z_2)$, $\underline{\rho}_i(t) = \sin^2(z_2)$, $\bar{\rho}_i(t) = 1 - \sin^2(z_2)$, $i = 1, 2, 3$.

By use of [Theorem 1](#) with $h = 0.05$, $\sigma = 50\%$, $\delta = 0.5$, $\underline{\xi} = 0.01$, $\bar{\xi} = 0.1$, $\gamma_1 = 0.9785$, $\gamma_2 = 0.8797$, $\gamma_3 = 0.6436$, and $\alpha = 5.358$, we have

$$K_1 = [-0.918 \ -6.3913], K_2 = [-0.0949 \ -6.4547], K_3 = [-0.0969 \ 6.6679].$$

Simulation results are shown in [Fig. 1](#), where system stability is demonstrated, and release instants and release intervals are exhibited under event-triggered mechanism.

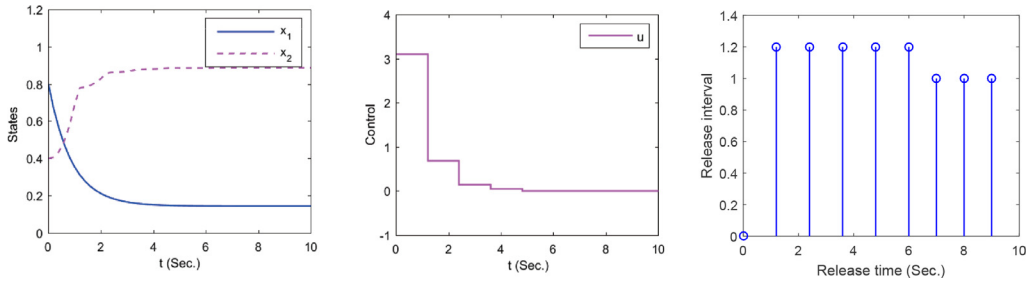


Fig. 1. State curves, control input, and release instants and release intervals.

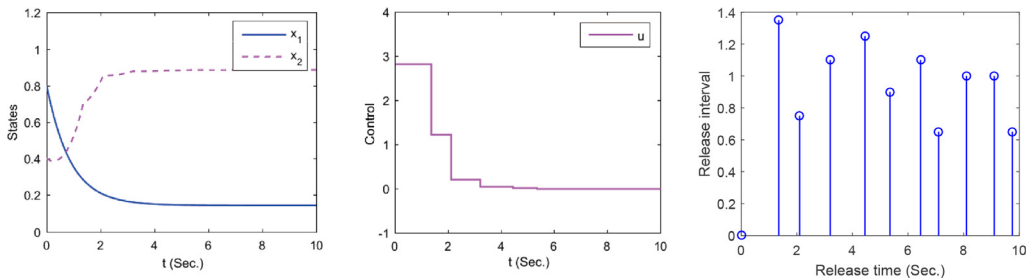


Fig. 2. State curves, control input, and release instants and release intervals.

Next, it is assumed that there do not exist missing data and transmission delay. Under [Corollary 1](#) with $h = 0.05$, $\delta = 0.5$, and $\alpha = 60.891$, an effective solution is solved as follows: $K_1 = [-0.1310 \quad -4.9189]$, $K_2 = [-0.1335 \quad -5.4294]$, $K_3 = [-0.1424 \quad -7.0851]$.

System states, the control input, and the event-triggered communication are displayed in [Fig. 2](#). As a result, the sampling number is reduced under the event-triggered control.

Obviously, the event-triggered control can reduce the sampling number and save the limited bandwidth.

Example 2. A one-link manipulator in [\[20\]](#)

$$\begin{aligned} \hat{D}\ddot{x} + \hat{B}\dot{x} + \hat{N}\sin(x) &= \psi, \\ \hat{M}\dot{\psi} + \hat{H}\psi &= u - \hat{L}\dot{x}, \end{aligned} \quad (41)$$

in which x denotes the position, $\dot{x}$ means the velocity, and ψ stands for the torque; and $\hat{D} = 1$, $\hat{B} = 1$, $\hat{H} = 1$, $\hat{M} = 0.1$, $\hat{L} = 0.2$, $\hat{N}$ is the uncertain parameter, and $x \in [-5\pi/12 \quad 5\pi/12]$.

Let $x(t) = [x_1(t) \quad x_2(t) \quad x_3(t)]^T = [x \quad \dot{x} \quad \psi]^T$. The one-link manipulator is modelled by the fuzzy system (2), where

$$\begin{aligned} A_1 &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & -2 & -10 \end{bmatrix}, B_1 = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & -2 & -10 \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}, \\ \vartheta_1^l(x_1(t)) &= \frac{1 - \sin(x_1(t))}{2.3}, \quad \vartheta_2^u(x_1(t)) = \frac{1 + \sin(x_1(t))}{1.7}, \quad \vartheta_1^u(x_1(t)) = \frac{1 - \sin(x_1(t))}{1.7}, \quad \vartheta_2^l(x_1(t)) = \frac{1 + \sin(x_1(t))}{2.3} \text{ with} \\ c_1(t) &= c_2(t) = \sin^2(x_1(t)), \quad \bar{c}_1(t) = \bar{c}_2(t) = 1 - \sin^2(x_1(t)) \quad \underline{\rho}_1(t) = \underline{\rho}_2(t) = \sin^2(x_1(t)), \\ \bar{\rho}_1(t) &= \bar{\rho}_2(t) = 1 - \sin^2(x_1(t)). \end{aligned}$$

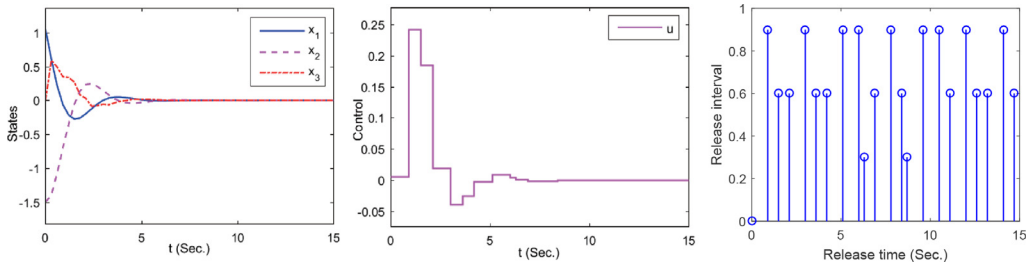


Fig. 3. State curves, control input, and release instants and release intervals with $\hat{N} = 2$.

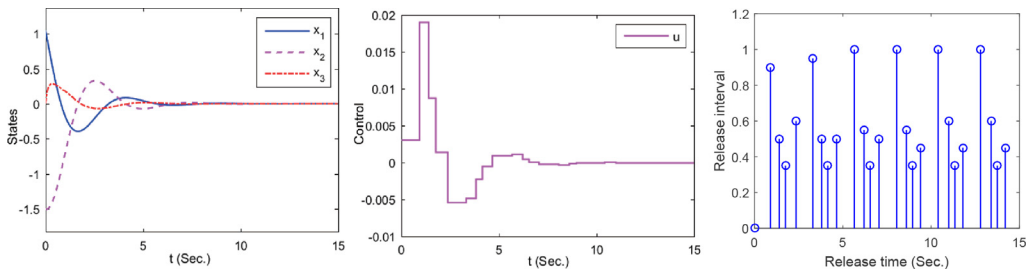


Fig. 4. State curves, control input, and release instants and release intervals with $\hat{N} = 2$.

Table 1
The number of transmitted data packets.

Methods	[12]	[25]	Corollary 1	Theorem 1
Packets	300	91	25	22

By use of Theorem 1 with $h = 0.05$, $\sigma = 50\%$, $\delta = 0.5$, $\underline{\xi} = 0.01$, $\bar{\xi} = 0.1$, $\gamma_1 = 0.85$, $\gamma_2 = 0.79$, and $\alpha = -65.861$, a feasible solution is obtained as follows:

$$K_1 = [-0.0005 \quad -0.0034 \quad 0.6726], \quad K_2 = [0.0004 \quad -0.0033 \quad 0.6987].$$

System states, the control input, and release intervals are displayed in Fig. 3. System stability is demonstrated. Obviously, the event-triggered control can reduce the sampling number and save the limited bandwidth. The implementation cost of the controller is further saved.

Next, it is assumed that there do not exist missing data and transmission delay. With Corollary 1 with $h = 0.05$, $\delta = 0.5$, and $\alpha = -15.367$, an effective solution is solved as follows:

$$K_1 = [-0.0015 \quad -0.0036 \quad 0.0799], \quad K_2 = [-0.0016 \quad -0.0031 \quad 0.0859].$$

Fig. 4 plots system states, control input, and release instants and release intervals, respectively. It can be seen that the sampling number is saved under the event-triggered mechanism.

For comparison, under the control methods in [12], [25] 1, and Theorem 1, the number of transmitted data packets is listed in Table 1.

It can be seen from Table 1 that when the simulation time is 15 seconds, data packets are transmitted 300 times and 91 times under the method [12] and [25], respectively. However, by using Corollary 1 and Theorem 1, the number of transmitted data packets are 25 and 22, respectively. So, the proposed event-triggered control can further reduce the sampling number and save the limited bandwidth. As a result, the design cost of the controller is decreased.

5. Conclusion

The IT2 fuzzy event-triggered control problem is concerned for NCSs. Using the sampled-state-error condition, an event-triggered controller is designed. It can be seen that the design can reduce the sampling number and save the limited bandwidth. However, the shortcoming of this design is that the states at $1.6h$, $3.7h$, $6.3h$ cannot be triggered. So, some useful information is neglected. In the following, event-triggered optimal control and sliding mode control [36,37] will be launched. Furthermore, this design can be extended to predictive control [27], asynchronous control [31], tracking control [38], synchronization control [39], stochastic control [40], and quantized control [41]. The plant can be expanded to strict-feedback systems [42] and large-scale systems [43].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Shandong Provincial Natural Science Foundation (ZR2021MF133) and the National Natural Science Foundation of China (61873071). This work of J. H. Park was supported by the [National Research Foundation of Korea](#) (NRF) grant funded by the Korea government (MSIT) (No. 2020R1A2B5B02002002).

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